PUSHFORWARD OF THE STRUCTURE SHEAF ALONG A UNIVERSALLY CLOSED MORPHISM IS INTEGRAL OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's reduction to a single section h and produces the monic equation explicitly; the properness and integrality inputs are applied inside the counted span.

Editorial note. The seven references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The factorization of a quasi-compact quasi-separated morphism through the relative spectrum of its pushforward. Remark remark-factorization-quasi-compact-quasi-separated, used at proof lines 10150-10155. Fixes the factorization that the lemma identifies with the normalization.
- Scheme theoretic image and its density for quasi-compact morphisms. Definition definition-scheme-theoretic-image and Lemma lemma-quasi-compact-scheme-theoretic-image, used at proof lines 10169-10175. Produces Z and the density of X -> Z.
- Permanence of universal closedness through a factorization. Lemma lemma-universally-closed-permanence, used at proof lines 10176-10177. Makes X -> Z closed, so that h(X) = Z.
- Images of proper morphisms are proper, and an affine universally closed morphism is integral. Lemmas lemma-image-proper-is-proper and lemma-integral-universally-closed, used at proof lines 10178-10182. Makes Z -> Y integral over an affine base.
- Permanence of quasi-compactness for the morphism h. Lemma lemma-quasi-compact-permanence, used at proof lines 10171-10173. Shows h is quasi-compact so that the scheme theoretic image argument applies.

Source reference. Original definition 082Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10268–10268. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08AJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10534–10534. Navigation pointer into the Stacks

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0415: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5280–5280. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4688–4688. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 082Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10269–10269. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6095–6095. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

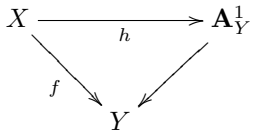
Source reference. Original remark 081X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10232–10232. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a quasi-compact, quasi-separated and universally closed morphisms of algebraic spaces over S. Then $f_*\mathcal{O}_X$ is integral over $\mathcal{O}_Y$. In other words, the normalization of Y in X is equal to the factorization*

$$X \longrightarrow \underline{\mathrm{Spec}}_Y(f_*\mathcal{O}_X) \longrightarrow Y$$

of Remark 081X.

Proof. The question is étale local on Y, hence we may reduce to the case where $Y = \mathrm{Spec}(R)$ is affine. Let $h \in \Gamma(X, \mathcal{O}_X)$. We have to show that h satisfies a monic equation over R. Think of h as a morphism as in the following commutative diagram



Let $Z \subset \mathbf{A}^1_Y$ be the scheme theoretic image of h, see Definition 082Y. The morphism h is quasi-compact as f is quasi-compact and $\mathbf{A}^1_Y \to Y$ is separated, see Lemma 03KS. By Lemma 082Z the morphism $X \to Z$ has dense image on underlying topological spaces. By Lemma 04NX the morphism $X \to Z$ is closed. Hence $h(X) = Z$ (set theoretically). Thus we can use Lemma 08AJ to conclude that $Z \to Y$ is universally closed (and even proper). Since $Z \subset \mathbf{A}^1_Y$, we see that $Z \to Y$ is affine and proper, hence integral by Lemma 0415. Writing $\mathbf{A}^1_Y = \mathrm{Spec}(R[T])$ we conclude that the ideal $I \subset R[T]$ of Z contains a monic polynomial $P(T) \in R[T]$. Hence $P(h) = 0$ and we win. □